

Regime-Gated PCA Microstructure Flow

A Cross-Sectional Residual-Flow Strategy on Crypto Perpetual Futures

QF635 Project Report

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1. Abstract

This report develops and evaluates a regime-gated, market-neutral, cross-sectional PCA residual-flow strategy on Binance USD-M perpetual futures, covering Bitcoin and nine altcoin perpetuals. The central research question is whether PCA-neutralized idiosyncratic taker-flow, funding crowdedness, and order-book confirmation can predict short-horizon cross-sectional returns after realistic trading costs.

The principal finding is that the signal is real but uneconomical to trade by taking liquidity. The PCA residual-flow signal exhibits a robust, statistically strong mean-reversion relationship with forward returns: the cross-sectional information coefficient (IC) is approximately -0.029 at the one-minute horizon ($t \approx -16$), and it decays consistently across horizons from 1 to 30 minutes over roughly 43,000 periods. Traded in the reversal direction and fully deployed, the signal produces a strong gross result of approximately $+6.5\%$ over thirty days, with a daily Sharpe ratio near 5.7 and a maximum drawdown of about 4%. However, the edge resides at the one-minute horizon and requires turning over the entire book every few minutes, so the break-even transaction cost is approximately 0.05 bp, far below any achievable Binance fee (maker ≈ 2 bp, taker ≈ 5 bp). Net of realistic costs the strategy loses heavily, and we therefore do not recommend real-capital deployment as a liquidity-taking strategy. The signal's natural application is liquidity provision, through market-making or quote-skew, or as a slower-horizon feature.

A secondary finding concerns the Bitcoin-trend regime gate. As specified, the gate is over-restrictive for a market-neutral book: it forces a no trades for roughly 80-90% of the test window and, when active, deploys during a poorly chosen 10% of bars, losing even at zero cost. A market-neutral cross-sectional strategy does not require a directional Bitcoin gate.

2. Motivation and Market Opportunity

Crypto perpetual futures offer rich, public microstructure data — taker-buy and taker-sell volume, funding rates, and Level-2 depth — alongside high retail participation, which makes short-horizon order-flow effects plausible. A complication is that common crypto-wide flow, in which nearly everything moves with Bitcoin, contaminates single-asset flow signals. Removing this common component via principal component analysis isolates idiosyncratic flow pressure, which is the candidate alpha. Funding crowdedness adds a slower positioning prior. Because the resulting opportunity is cross-sectional and traded long-short, the book is market-neutral and avoids directional crypto exposure.

3. Market Microstructure Analysis

The core flow primitive is the Taker-Flow Imbalance (TFI), defined per asset and per minute as

$$\frac{(\text{TakerBuy} - \text{TakerSell})}{(\text{TakerBuy} + \text{TakerSell})}$$

The common factor is estimated by a rolling principal component analysis over a 120-bar window on the cross-section of TFI: the first principal component approximates broad crypto-wide flow, while the second captures secondary rotation. The first component explains only a modest share of variance — about 14% on the test data — indicating that idiosyncratic flow is substantial.

In live and paper trading, several order-book features supplement the flow primitive: a weighted order-book imbalance (with weights $w_k = \frac{1}{k}$), microprice deviation, spread in basis points, a rolling-median spread filter, and an ADV20 liquidity filter. An important caveat is that Binance provides no historical Level-2 books over its REST interface, so the backtest synthesizes a neutral book from klines; this limitation is discussed in Sections 8 and 14.

4. Strategy Design

The strategy comprises three layers, following the canonical specification.

1. BTC regime gate (daily). Five-day and twenty-day momentum, together with twenty-day realized volatility measured against its Q80(252) threshold, classify the regime as Strong (1.0× gross), Moderate (0.5×), or Risk Off (0×).
2. PCA residual-flow alpha (one-minute). A rolling PCA feeds an OLS residualization of TFI on the first two principal components, using a 60-bar beta, followed by a cross-sectional z-score. The alpha blends $0.8 \times Z(\text{Residual})$ with $0.2 \times \text{Funding Signal}$, and direction is taken from out-of-sample IC, with an optional stability gate.
3. Microstructure confirmation and portfolio. A composite MicroScore = $0.4 \times \text{OBI} + 0.3 \times \text{MicroSignal} + 0.3 \times \text{TFI_short}$ drives Q80/Q20 selection over the liquidity-eligible non-BTC universe. Positions are volatility-scaled and market-neutral, subject to a single-asset cap, and rebalanced every five minutes.

5. Model Selection and Alternatives

PCA residualization is preferred over raw TFI because raw flow is dominated by the common factor; residualizing isolates the cross-sectional bet and enforces market neutrality. Principal-component counts of one, two, and three were tested.

Cross-sectional z-scores combined with Q80/Q20 selection are preferred over a fixed threshold because rank and quantile selection are robust to scale drift and are naturally market-neutral.

Rather than assuming continuation, the trading direction is set from realized IC. We compare a simple rule based on the sign of IC(1) against a stricter IC stability gate that combines the t-statistic with cross-horizon sign stability. The data show the sign is negative — that is, mean-reverting — so a hardcoded continuation prior would be wrong, as detailed in Section 9.

One alternative was considered but not pursued as the primary engine: Avellaneda–Stoikov market-making. It is in fact the natural home for this fast signal, as discussed in Section 15, but it represents a different execution model.

6. Implementation

The system is a single flat Python package with an event-driven pipeline: feed, book, strategy, risk, broker, portfolio, and record, supplemented by a Streamlit dashboard and a historical backtester. A single shared

alpha engine, spanning `pca_flow.py`, `microstructure_confirmation.py`, and `combined.py`, is used identically by the live framework, the backtester, and the user interface. Live execution, through `broker.LiveBroker` and `binance_rest.py`, issues signed REST orders with tick and step rounding and GTX post-only handling, an asynchronous user-data fill stream, and cancel/replace logic. A suite of 69 unit tests covers the broker, risk, portfolio, runner, PCA, microstructure, funding, live-broker parsing, and the strategy overlays.

7. Risk Management

Pre-trade gates in `risk.py` enforce a per-order quantity cap, a total-exposure cap sized by binary search, a per-symbol rate limit, and reduce-only handling. Circuit breakers include a daily-loss HALT at -3% and a consecutive-failure HALT. At the portfolio level the book is market-neutral ($\Sigma w = 0$), with a single-asset cap and regime-scaled gross exposure. In live operation, the system fails fast in a kill-switch style on credential or venue errors, and funding is settled into PnL.

8. Backtesting Methodology

The backtest is an event-by-event replay of the canonical strategy over real Binance one-minute Klines, daily Bitcoin candles, and funding history, with look-ahead protection: the regime uses only completed prior daily candles, and the PCA fits on past windows only. Profits and losses are computed on a weights basis — target weights are assumed achieved each bar, and turnover is charged a configurable per-unit cost via `--cost-bps` — with funding applied at settlement timestamps.

A material limitation is the absence of historical Level-2 data: books are synthesized from Klines with neutral OBI and microprice. The microstructure-confirmation layer is therefore only approximated in backtest, and its true value must be assessed live or on paper; results are reported with confirmation both enabled and disabled in Section 11. The cleanest test of predictive power is the signal diagnostic, the per-period cross-sectional Spearman IC of the raw residual-flow signal against forward returns, which is independent of gate, cost, and regime. Runs are reproduced with

```
python backtest.py --start 2026-05-19 --days 30
```

, and committed output is stored in `results/`.

9. Performance Analysis

Results below cover the window 19 May 2026 to 18 June 2026, comprising 43,200 one-minute bars. Table 1 reports the information coefficient of residual flow against forward returns over 43,020 periods.

Horizon	IC	t-stat	Hit %
1 min	-0.0288	-16.6	45.9
2 min	-0.0267	-15.5	46.1
5 min	-0.0242	-14.2	46.4
10 min	-0.0184	-10.8	47.0

Horizon	IC	t-stat	Hit %
30 min	-0.0141	-8.3	47.7

Table 1. Information coefficient of the residual-flow signal versus forward returns.

The IC is robust and negative, indicating mean-reversion, decays smoothly with horizon, and holds the same sign across all horizons — the signature of a real signal. Table 2 summarizes the profit and loss of the full strategy with all gates active, as specified.

Metric	Value
Total return (30d)	-0.50%
Annualized return	-5.91%
Annualized volatility	0.97%
Sharpe (annualized, from daily P&L)	-5.03
Max drawdown	-0.53%
Win rate (in-market bars)	43.0%
Rebalances / avg turnover / total turnover	122 / 0.351 / ≈ 42.8
Total trading cost	0.43%
Time in market	1.0%
Regime mix	RiskOff 90%, Moderate 10%, Strong 0%
Benchmark (equal-weight buy-hold, 9 alts)	-18.76%
Excess return vs benchmark	+18.26%

Table 2. Full-strategy performance, all gates enabled.

The full strategy barely trades — the regime and IC gates are restrictive — and loses slightly to costs in absolute terms. Because the book is continuously weighted and market-neutral, the meaningful activity measures are the number of rebalances (122) and turnover (total $\approx 42.8\times$ the book) rather than a discrete trade count. The most striking feature is the benchmark comparison in Section 10: a passive equal-weight long of the same nine altcoins lost 18.76% over this window, so the market-neutral book's -0.50% represents 18.26% of downside protection. This is capital preservation — the regime gate held the book roughly flat through the sell-off — rather than captured alpha. The window is therefore inconclusive on the alpha itself, which motivates full deployment in Section 11 to measure the edge directly.

10. Benchmark Comparison

Because the strategy is market-neutral ($\Sigma w = 0$), it is reported against two baselines, shown in Table 3.

Benchmark	Return (30d)	Strategy (-0.50%) vs benchmark
Cash / risk-free ($\sim 0\%$)	$\approx 0.00\%$	-0.50% (underperforms)
Equal-weight buy-hold (9 alts, long-only)	-18.76%	+18.26% (outperforms)

Table 3. Strategy return relative to two baselines.

Two honest readings are simultaneously true. Against cash, the as-specified strategy loses 0.50%; the fully deployed gross alpha of +6.5% (Section 11) beats cash before costs but not after, so by the cash yardstick the strategy is not yet investable. Against the long-only altcoin basket, which fell 18.76% over the window — the regime gate read RiskOff 90% of the time for good reason — the market-neutral book outperformed by 18.26%. This reflects the value of market-neutrality combined with the regime gate, but it is downside avoidance rather than alpha extracted from the signal; the book was roughly flat and cash-like during the drawdown, and a passive short of the basket would have done better still. We therefore do not claim the 18.26% as strategy skill, and cash remains the fair absolute benchmark for a neutral book. The benchmark is computed by the backtester's `compute_metrics` routine as an equal-weight buy-hold of the traded universe over the identical window.

11. Robustness and Ablations

Table 4 backs out gross performance from net plus cost over the thirty-day window, trading in the reversal direction wherever deployed.

Configuration	Time in mkt	Gross	Cost @1bp	Net @1bp
Full strategy (regime + IC gate on)	1.0%	~+0.0%	0.4%	−0.50%
No regime, reversal, no confirm, cost 0	99.6%	+6.49%	0	+6.49%
No regime, reversal, no confirm, cost 1bp	99.6%	+6.49%	~126%	≈ −120%
+ rebalance 30, smooth 15	99.6%	+0.50%	19.3%	−18.84%
+ rebalance 60, smooth 1 (fresh)	99.6%	+2.63%	10.7%	−8.11%
+ rebalance 60, smooth 30	99.6%	−1.29%	9.0%	−10.31%

Table 4. Ablation of gates, rebalance frequency, and smoothing.

Removing the regime gate unlocks the real edge, raising gross Sharpe to roughly 5.7 at zero cost, but turnover is the binding constraint: the break-even cost is approximately 0.05 bp for the fast configuration, improving to about 0.24 bp at hourly rebalancing, still far below realistic fees. Critically, smoothing or slowing the strategy to reduce turnover destroys the gross edge, because the edge is the fast reversion itself. Both the no-trade band and the confirmation gate reduce trades without rescuing net return.

Cross-window stability is a concern. The one-minute IC is −0.029 in May–June 2026 but only −0.010 in March 2026, within the noise floor, so the effect is regime- and period-dependent. In any live use the IC stability gate should remain enabled rather than hardcoding the sign.

12. Optimization and Parameter Impact

Direction has the largest impact: continuation (+1) is wrong and reversal (−1) is correct, which flips the sign of P&L. Removing the regime gate is the single largest improvement, raising deployment from 1.4% to 99.6% of bars and capturing the gross edge; the Bitcoin-trend gate is inappropriate for a neutral book. Turnover controls — rebalance frequency, smoothing, and the no-trade band — reduce cost but reduce gross roughly proportionally, and no setting clears the fee wall. The alpha blend, PCA windows, and

quantile thresholds are secondary and are not worth tuning until the cost problem is solved, since tuning a cost-dominated strategy overfits noise.

13. Production Deployment

The live path is implemented and tested against the Binance API shape, including asynchronous market-data and user-data streams, signed REST orders with tick and step rounding and GTX post-only handling, cancel/replace, listenKey keepalive, fills booked into the portfolio, a kill switch, and testnet operation by default. The system is not production-proven: it should be dry-run on testnet first, and per-symbol exposure caps, reconciliation hardening, and monitoring should be added before any real capital is committed.

14. Limitations

1. Uneconomical as a liquidity-taker. The break-even cost of approximately 0.05 bp is far below real fees; this is the dominant limitation.
2. Microstructure layer not validated in backtest. Synthetic books, in the absence of historical Level-2 data, mean its real contribution is unmeasured; claims about order-book confirmation apply to live and paper trading only.
3. Single short window and a regime-dependent signal. The analysis rests on one thirty-day sample, and the IC sign varies by period.
4. Weights-based backtest. There is no fill simulation, no partial fills, and no queue modeling, so net results are an approximation.
5. Probe orders and the queue-aware fill model are designed but not implemented.
6. Benchmark caveat. The +18.26% versus the long-only altcoin basket is downside avoidance — a neutral, cash-like book during an 18.76% sell-off — not extracted alpha. Cash (~0%) is the fair absolute baseline for a market-neutral strategy, and against cash the as-specified strategy loses.

15. Conclusion

Real capital should not be deployed as specified. The research produced a genuine, economically sensible signal: idiosyncratic taker-flow mean-reverts intraday. However, it is too fast to monetize by crossing the spread at any realistic fee, and the regime gate is mis-specified for a market-neutral book. Two defensible paths forward follow. The first is to reframe the signal as a market-making or quote-skew input that earns the spread rather than paying it — the natural home for a fast, correct micro-signal. The second is to develop a slower-horizon feature with comparable IC and far lower turnover. The implemented system provides a solid research and paper-trading platform for that next step.